1 Oct 20-22

1,2,4,6,7

1:$\Delta\nu = \frac{c}{2\mu L} = \frac{3*10^8}{1} = 3*10^8$ Bec : $\Delta\nu = 6*10^8$

So : $n = \frac{\Delta\nu}{3*10^8}$=2.

q could be : q-1;q;q+1.

and $\nu = \frac{qc}{2\mu L}$ so $q = \frac{2\mu L\nu}{c} = \frac{5.85*10^{14}}{3*10^8} = 1.95*10^{16}$and q+1 q-1.

2:

$\Delta\nu = \frac{3*10^8}{2} = 1.5*10^8$=0.1 $*\Delta\nu$

so n=10 ,q could have 11 . if q only neede 1 ,so

$\Delta\nu = 1.5*10^9 = \frac{3*10^8}{2L}$so L=0.1 m .

4:

(1):$P/S = \frac{50W}{(25\mu m)^2\pi}\mu m^2$= K

(2):$\frac{K}{10^4}$=N_1;$\frac{K}{10^3}$=N_2

6:

$\theta = \frac{1.22*632.8nm}{2mm} = \theta_1$

θ_0=$\frac{\lambda}{\pi\omega_0}$=$\frac{\lambda}{\pi\sqrt{\frac{\lambda L}{2\pi}}}$

7:

ω_0=$\sqrt{\frac{\lambda L}{2\pi}}$

$\omega(z)$=$\omega_0\sqrt{1+\frac{\lambda z}{\pi\omega_0^2}^2}$

$L = 0.4; \lambda = 0.6328\mu m; z = 56cm$

9:

$R = z[1+(\frac{f}{z})^2]\frac{3\sqrt{3}}{7} = f, R_1' = R_2' = \frac{6\sqrt{3}}{7}$

$\omega_0 = \sqrt{\frac{\lambda f}{\pi}} = \sqrt{\frac{\lambda 3\sqrt{3}}{7\pi}}$

$\omega = \omega_0\sqrt{1+(\frac{f}{z})^2} = \sqrt{\frac{\lambda 3\sqrt{3}}{7\pi}}\sqrt{1+(\frac{3\sqrt{3}}{7z})^2}$

$\frac{R(z_1)-z_1}{z_1} = (\frac{f}{z_1})^2; \frac{R(z_2)-z_2}{z_2} = (\frac{f}{z_2})^2$

$\frac{1.5-z_1}{3+z_1} = \frac{1-z_1}{z_1}; z_1+z_2 = 1; \frac{6}{7} = z_1; z_2 = \frac{1}{7};$

10:

$2 = 1[1+f^2]$;f=1;$\omega = \omega_0\sqrt{1+(\frac{f}{z})^2} = \sqrt{\frac{2\lambda}{\pi}} = 2.5977mm$

$\omega = \omega_0\sqrt{1+(\frac{f}{z})^2} = \omega_0\sqrt{1+f^2} = \omega_0\sqrt{R} = \sqrt{\frac{\lambda\sqrt{R-1}R}{\pi}} = 0.3$

$0.3^2\pi/\lambda = R\sqrt{R-1}$;R=41.7733cm.

$\omega_0 = \sqrt{\frac{\lambda\sqrt{R}}{\pi}}$

$\omega = \omega_0\sqrt{1+f^2}; R = 1+f^2; f = \sqrt{R-1}; \omega_0\sqrt{R} = \sqrt{\frac{\lambda R\sqrt{R}}{\pi}}$

$R = (0.3^2\pi/\lambda)^{2/3} =$

$f = \frac{\sqrt{2(R-2)^2(2R-2)}}{2R-4} = \frac{2\sqrt{R-1}}{2} = \sqrt{R-1}$

$\omega_0 = \sqrt{\frac{\lambda\sqrt{R}z}{\pi}\frac{R}{z}}; f = \sqrt{(\omega_0^2\pi/\lambda)^{2/3}}$

$(\omega_0^2\pi/\lambda)^{2/3} = R$

$\omega = \omega_0\sqrt{1+(\omega_0^2\pi/\lambda)^{2/3}}$

2 redo HW all

redo

2.1 chapter 1

2:

(1):

$$\frac{n_2}{n_1} = \frac{g_2}{g_1} e^{\frac{-h\nu}{kT}}$$
$$= e^{-\frac{h\times 3\times 10^9}{1.38\times 10^{-23}\times 300}} = 0.995$$

(2)

$$0.1 = e^{-\frac{h\frac{c}{\lambda}}{1.38\times 10^{-23}T}}$$
$$-ln10 = -\frac{h\frac{c}{\lambda}}{1.38\times 10^{-23}T}$$
$$T = \frac{h\frac{c}{\lambda}}{1.38\times 10^{-23}ln10} = 6259k$$

3:

(1):

$$\frac{n_2}{n_1} = \frac{g_2}{g_1} e^{-\frac{E}{kT}}$$
$$n_2 = n_1\frac{g_2}{g_1} e^{-\frac{E}{kT}} = 10^{20}\times 4\times e^{-\frac{1.64\times 10^{-18}}{1.38\times 10^{-34}\times 2700}} = 30.66$$

(2):

$$P = I\times A = h\nu\times n = 1.64\times 10^{-18}\times 10^8\times n_2 = 4.92\times 10^{-9}$$

4:

(1):

$$\frac{q}{q_n} = \frac{B_{21}\rho}{A_{21}} = \frac{c^3\rho}{8\pi h\nu^3} = \frac{1}{2000}$$
$$\rho = \frac{8\pi h\nu^3}{2000 \times c^3} = \frac{8\pi h}{2000 \times \lambda^3} = 3.85 \times 10^{-17}$$

(2):

$$\frac{q}{q_s} = \frac{c^3\rho}{8\pi h\nu^3} = \frac{\lambda^3\rho}{8\pi h} = 7.603 \times 10^9$$

11:

$$\nu = \nu_0(1 + \frac{v}{c}); \nu_{0.1} = \nu_0 \times 1.1; \nu_{-0.1} = \nu_0 \times 0.9$$
$$\nu_{0.5} = \nu_0 \times 1.5; \nu_{-0.5} = \nu_0 \times 0.5$$
$$\nu_0 = 4.74 \times 10^{14}$$
$$\nu_{0.1} = 5.214 \times 10^{14}; \nu_{-0.1} = 4.266 \times 10^{14};$$
$$\nu_{0.5} = 7.11 \times 10^{14}; \nu_{-0.5} = 2.370 \times 10^{14}$$

12:

$$\delta\nu = \nu - \nu_0 = \nu_0 \times (1 + \frac{v}{c}) - \nu_0 = \frac{v}{c} \times \nu_0 = 8.848 \times 10^8 Hz$$

13:
(1):

$$I = I_0 \times e^{-Gz}; \frac{I}{I_0} = e^{-Gz} = e^{-0.01 \times 10^3 \times 10 \times 10^{-2}} = e^{-1} = 36.78\%$$

(2):

$$2I = Ie^{G \times 1m}; ln2 = G \times 1; G = 0.6931$$

2.2 chapter 2

1:

$$G = \Delta h\nu/cf(\nu)B_{21} = 5\times 10^{24}\times f(\nu)\frac{A_{21}c^2}{8\pi\nu^2\mu^2} = 5\times 10^{24}\frac{1}{2\times 10^{11}}\frac{\lambda^2}{8\pi\tau\mu^2}$$
$$= 71.037$$

2:

$$n_1+n_2 = 10^{18}; n_2 = 4n_1; n_1 = 2\times 10^{17}; n_2 = 8\times 10^{17}; \Delta n = n_2 - n_1\frac{g_2}{g_1} =$$
$$(8\times -2\times\frac{3}{5})\times 10^{17} = \frac{34}{5}\times 10^{17}$$
$$g = \Delta n\frac{h\nu}{c}f(\nu)B_{21} = \frac{34}{5}\times 10^{17}\frac{h\nu}{c}f(\nu)A_{21}\frac{c^3}{8\pi h\nu^3} = \frac{34}{5}\times 10^{17}f(\nu)\frac{c^2}{8\pi\nu^2\tau}$$
$$= 72.228$$

3:
(a)

$$0 < (1-\frac{L}{R})^2 < 1$$
$$-1 < 1-\frac{L}{R} < 1 : \frac{L}{R} > 0; \frac{L}{R} < 2, R > 0, R > \frac{L}{2}$$
$$R > 30cm$$

(b):

$$0 < (1-\frac{L}{4L})(1-\frac{L}{R}) < 1; 0 < \frac{3}{4}(1-\frac{L}{R}) < 1$$
$$\frac{L}{R} < 1; -\frac{1}{3} < \frac{L}{R}$$
$$R > 0 : R > L; R > -3L$$
$$R < 0 : R < L : R < -3L$$
$$R : R > L | R < -3L$$

7:

$$
\begin{aligned}
&G = \frac{(\frac{\Delta\nu}{2})^2 G_0(\nu_0)}{(\nu-\nu_0)^2 + (1+\frac{I}{I_s})(\frac{\Delta\nu}{2})^2} \\
&\text{if:}(\nu-\nu_0)^2 >> (1+\frac{I}{I_s})(\frac{\Delta\nu}{2})^2 \\
&|\nu-\nu_0| = \Delta\nu' = \sqrt{1+\frac{I}{I_s}}\Delta\nu \\
&I = I_s; \Delta\nu' = \sqrt{2}\Delta\nu
\end{aligned}
$$

8:

$$
\begin{aligned}
\sigma_e(\nu) &= \frac{G(\nu)}{\Delta n} = \frac{\Delta n \frac{h\nu}{c} f(\nu) B_{21}}{\Delta n} = \frac{h\nu f(\nu) B_{21}}{c} \\
&= \frac{f(\nu) A_{21} c^2}{8\pi\nu^2} = \frac{f(\nu)c^2}{8\pi\mu^2\tau\nu^2}
\end{aligned}
$$

10:

$$
\begin{aligned}
&(a)G_D(\nu) = \frac{G_D(\nu_1)}{\sqrt{1+\frac{I}{I_s}}} = \frac{3\times 10^{-4}}{10^{-3}\times\sqrt{1+\frac{5}{3}}} = 0.1837 \\
&(b)a_a = a_i - \frac{1}{2L} ln(r_1 r_2) = G \\
&\quad a_i - G = \frac{1}{2L} ln(r_1 r_2) = \frac{1}{L} ln(r) = (a_i - G) \\
&\quad = r = e^{L(a_i - G)} = 0.9906 \\
&(c)P = tIA = 0.008 \times 50 \times 10^4 \times 0.11 \times 10^{-6} = 4.4 \times 10^{-4} \\
&\quad = 44mW
\end{aligned}
$$

11:

$$
\begin{aligned}
&G = \Delta n h \frac{\nu}{c} f(\nu) B_{21} \\
&\Delta n = \frac{cG}{h\nu f(\nu) B_{21}} = \frac{8\pi\nu^2 a_a \tau}{f(\nu)c^2} \\
&R = 1 - 0.1 \times a_a = 0.9833; 9.833 = 10 - a_a \\
&a_a = 0.167 \\
&\Delta n = 1.048 \times 10^{15}
\end{aligned}
$$

2.3 chapter 3

1:

$$\Delta\phi = \frac{c}{2\mu L} = 3 \times 10^8$$
$$\frac{\Delta\nu}{\Delta\phi} = 2; N_q = 3$$
$$\phi = 5.85 \times 10^{14} = \frac{qc}{2\mu L} = q \times 3 \times 10^8$$
$$q = 1.95 \times 10^6; q+1 = 1.95 \times 10^6 + 1; q-1 = 1.95 \times 10^6 - 1$$

2:

$$\Delta\phi = \frac{c}{2\mu L} = 1.5 \times 10^8$$
$$\frac{\Delta\nu}{\Delta\phi} = 10; N_q = 11$$
$$\text{if:} N_q = 1; \Delta\phi > \Delta\nu : \frac{c}{2\mu L} = \frac{1.5 \times 10^8}{L} > 1.5 \times 10^9$$
$$L < 0.1m$$

4:
(1):

$$p/s = \frac{50}{s}; s = \frac{\pi d^2}{4} = 1.963 \times 10^{-9}$$
$$p/s = 2.54 \times 10^{10} W/m^2 = 2.54 \times 10^6 W/cm^2$$

(2):

$$254\times$$
$$2540\times$$

6:

$$\theta_0 = \frac{\lambda}{\pi\omega_0} = \frac{\lambda}{\pi}\sqrt{\frac{\pi}{\lambda f}} = \sqrt{\frac{\lambda}{\pi f}} = 1.1588 \times 10^{-3}$$
$$\theta_r = \frac{1.22 \times 632 \times 10^{-9}}{2 \times 10^{-3}} = 3.855 \times 10^{-4}$$

7:

$$\omega = \omega_0\sqrt{1 + (\frac{z}{f})^2}; \omega_0 = \sqrt{\frac{\lambda f}{\pi}} = 2 \times 10^{-4}$$
$$\omega = 2 \times 10^{-4} \times \sqrt{1 + (\frac{0.56}{0.2})^2} = 5.94 \times 10^{-4}$$

9:

$$R_1 z_1 = z_1^2 + f^2; R_2 z_2 = z_2^2 + f^2$$
$$R_1 z_1 - z_1^2 = R_2 z_2 - z_2^2; z_1 + z_2 = 1$$
$$R_1 z_1 - z_1^2 = R_2(1 - z_1) - (1 - z_1)^2; R_1 = 1.5, R_2 = 4$$
$$1.5 z_1 - z_1^2 = 4 - 4z_1 - 1 - z_1^2 + 2z_1; 3.5 z_1 = 3; z_1 = \frac{6}{7}; z_2 = \frac{1}{7}$$
$$f^2 = \frac{4}{7} - \frac{1}{49} = \frac{27}{49}; f = \frac{3\sqrt{3}}{7}$$
$$\omega_0 = \sqrt{\frac{\lambda f}{\pi}} = 3.486 \times 10^{-4}$$
$$\omega_2 = \omega_0\sqrt{1 + (\frac{z_2}{f})^2} = 3.549 \times 10^{-4}$$
$$\omega_1 = \omega_0\sqrt{1 + (\frac{z_1}{f})^2} = 5.32 \times 10^{-4}$$

10:

$$
\begin{aligned}
&(a)\omega_0 = \sqrt{\frac{\lambda f}{\pi}} = 1.836 \times 10^{-3} \\
&\quad \omega = \omega_0 \sqrt{1 + (\frac{z}{f})^2} = 2.597 \times 10^{-3} \\
&(b)\omega = \omega_0 \sqrt{1 + (\frac{z}{f})^2} = \sqrt{\frac{f\lambda}{\pi}} \sqrt{1 + (\frac{z}{f})^2} = 0.3 \times 10^{-2} \\
&\quad \text{solve f :} f = 2.2161 \\
&\quad R = z + \frac{f}{z}^2 = 1 + f^2 = 5.911
\end{aligned}
$$

2.4 chapter 4

1:

$$
\begin{aligned}
\Delta\phi &= \frac{c}{2\mu(L_2 + L_3)}; N_q = 3, \frac{\Delta\nu}{\Delta\phi} = 2 \\
750 \times 10^6 &= \frac{3 \times 10^8}{2\mu(L_2 + L_3)}; (L_2 + L_3) = \frac{1}{5} = 0.2
\end{aligned}
$$

2:

$$
\begin{aligned}
&(1)\omega_{00} = \sqrt{\frac{\lambda f}{\pi}} = 1.419 \times 10^{-4} \text{add a hole at mirror surface} \\
&\quad \omega = \omega_0 \sqrt{1 + (\frac{z}{f})^2} = 2.007 \times 10^{-4} \\
&(2)\omega_{11} = \sqrt{2+1}\omega_{00} = 3.476 \times 10^{-4}
\end{aligned}
$$

3:

$$
\begin{aligned}
&\omega_0 = \sqrt{\frac{f\lambda}{\pi}}; f = \omega_0^2 \pi / \lambda = 0.1985 \\
&q = 0.6 + 0.1985i; T = \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix} \\
&q' = \frac{q}{-\frac{100}{3} \times (0.6 + 0.1985i) + 1} = -0.0314 + 4.903 \times 10^{-4} i
\end{aligned}
$$

$$\omega' = \sqrt{\frac{\lambda f'}{\pi}} = 9.9 \times 10^{-6}$$

4:

$$\theta_1 = \frac{\lambda}{\pi\omega_1} = 1.007 \times 10^{-3}$$
$$\theta_2 = \frac{\lambda}{\pi\omega_2} = 4.026 \times 10^{-3}$$

4 times bigger

5:

$$M = \frac{f_2\omega}{f_1\omega_0} = \sqrt{2} \times \frac{f_2}{f_1} = 11.31$$

8:

$$(1)\Delta\phi = \frac{c}{2\mu L} = \frac{c}{2 \times 1.5} = 10^8$$
$$T = 10^{-8}s$$
$$\tau = \frac{T}{N_q} = \frac{10^{-8}}{1000} = 10^{-11}$$
$$P_{peak} = NP_{aver} = 1000 \times 1 = 10^3$$
$$(2)T = \frac{2\pi}{\omega} = \frac{2\pi}{2\pi f} = \frac{1}{f} = 10^{-8}$$
$$f = 10^8$$

9:

$$N_q = \frac{\Delta\nu}{\Delta\phi} + 1 = \frac{7.5 \times 10^{12}}{\frac{c}{2(1.52\times0.2+0.1)}} + 1 = 20201$$

$P_{peak} = N_q \times P_{aver}$;

20201 times bigger